ADDIS ABABA INSTITUTE OF TECHNOLOGY

Software engineering department

Mobile Application Development (SECT-3113)

Project Title: Networking, REST APIs & Data Handling in Flutter

ASSIGNMENT 2

Recipe Browser – Flutter App

GITHUB Repository Link: https://github.com/Tamrat16/Mobile-Application-Development-ASSIGNMENT-2/

Name Id

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Submitted to**: Abel Tadesse**

Submissin date: May 3 , 2026

Submission Checklist

Code & Architecture

- ☑ http package used for all requests (not Dio)
- ☑ All URIs built with Uri.https() or Uri.parse() — no string concatenation
- ☑ 10-second timeout on every request
- ☑ All required model classes present with fromJson, toJson, copyWith
- ☑ All model fields are final and explicitly typed (no dynamic)
- ☑ Dedicated API service class in lib/services/
- ☑ No HTTP imports in any Screen or Widget file
- ☑ _checkResponse() method throws ApiException for non-200 responses
- ☑ FutureBuilder handles all 4 states (waiting, error, no-data, data)
- ☑ Retry button works when network connection is restored

Error Handling

- ☑ SocketException caught and shows 'No internet connection'
- ☑ TimeoutException caught and shows 'Request timed out'
- ☑ Non-200 status codes caught by ApiException
- ☑ FormatException caught for malformed JSON
- ☑ Generic catch-all Exception handler present

Project Structure

- ☑ lib/models/ folder with one file per model
- ☑ lib/services/ folder with service class and api_exception.dart
- ☑ lib/screens/ folder with all screen files
- ☑ All files named in snake_case

Submission

- ☑ GitHub repository is public
- ☑ At least 5 commits with meaningful messages
- ☑ README.md includes all 6 required items (name, ID, track, description, setup, limitations)
- ☑ No API keys committed to the repository
- ☑ .env file in .gitignore (if applicable)
- ☑ Screen recording shows: loading, search, detail, error state, retry
- ☑ Google Classroom submission form submitted with correct GitHub URL

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